

Practical Workbook
CT-265
Introduction to Tools and Techniques for
Data Science



Name: Moazzam Farooqui

Roll No: CT-24068

Batch: 2024

Department: CSIT

Department of Computer Science & Information Technology
NED University of Engineering & Technology
Table of Contents

S. No.	Object	Date
01	To build a strong foundation in Python programming by understanding the basic syntax, data types, operators, control flow, functions, and commonly used data structures.	12-9-2026
02	Learn how to use Pandas to make and work with Data Frames.	12-9-2026
03	To understand the NumPy library, which provides support for large, multi-dimensional arrays and matrices.	
04	To create effective static visualizations for data exploration and insight extraction.	
05	To apply foundational probability and statistics concepts to understand datasets.	
06	To provide a hands-on introduction to applying basic machine learning models for prediction.	
07	To acquire data from the web using web scraping techniques and extract structured information from websites.	
08	To apply natural language processing techniques to analyze and extract insights from customer review data.	
09	To understand the fundamentals of Microsoft Power BI for importing data, transforming it, building data models, creating visualizations, and designing interactive dashboards.	
10	To learn the basics of Tableau for connecting data, creating worksheets, applying filters, building dashboards, and presenting analytical stories from data.	

LAB # 02 DATA PROCESSING WITH PANDAS

EXERCISE # 01: DATAFRAME CREATION AND INSPECTION

Q1. Create a DataFrame df_products for an online store with the following columns: Product_ID, Product_Name, Price, Category, Stock_Quantity.

SOLUTION:

```
1  import pandas as pd
2
3  data={
4      "Product_ID":[1,2,3,4,5],
5      "Product_Name":["Shampoo","Hair Brush","Soap","Charger","Tooth Brush"],
6      "Price":[120000,2500,4500,6000,35000],
7      "Category":["Hygiene","Personal Accesories","Hygiene","Electronics","Hygiene"],
8      "Stock_Quantity":[1000,500,10000,1000,1500]
9  }
10
11  df=pd.DataFrame(data)
12
13  print(df)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#01Q1.py
  Product_ID Product_Name  Price  Category  Stock_Quantity
0          1        Laptop 120000  Electronics           10
1          2         Mouse   2500  Accessories           50
2          3      Keyboard   4500  Accessories           30
3          4   Headphones   6000        Audio           20
4          5        Monitor  35000  Electronics          15
```

Q2. Add at least 5 rows of sample data.

NOTE: 5 rows were already added in Q1

SOLUTION:

```
1  import pandas as pd
2
3  data={
4      "Product_ID": [1,2,3,4,5],
5      "Product_Name": ["Shampoo", "Hair Brush", "Soap", "Charger", "Tooth Brush"],
6      "Price": [120000, 2500, 4500, 6000, 35000],
7      "Category": ["Hygiene", "Personal Accesories", "Hygiene", "Electronics", "Hygiene"],
8      "Stock_Quantity": [1000, 500, 10000, 1000, 1500]
9  }
10
11 df=pd.DataFrame(data)
12
13 print(df)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#01Q1.py
  Product_ID Product_Name  Price  Category  Stock_Quantity
0          1      Laptop 120000  Electronics           10
1          2        Mouse   2500  Accessories           50
2          3    Keyboard   4500  Accessories           30
3          4  Headphones   6000        Audio           20
4          5      Monitor  35000  Electronics           15
```

Q3. Use the .info() and .describe() methods to inspect your DataFrame. What is the average price? What data type is the Category column?

SOLUTION:

```
1 import pandas as pd
2
3 data={
4     "Product_ID":[1,2,3,4,5],
5     "Product_Name":["Shampoo","Hair Brush","Soap","Charger","Tooth Brush"],
6     "Price":[120000,2500,4500,6000,35000],
7     "Category":["Hygiene","Personal Accesories","Hygiene","Electronics","Hygiene"],
8     "Stock_Quantity":[1000,500,10000,1000,1500]
9 }
10
11 df=pd.DataFrame(data)
12 print(df)
13
14 df.info()
15 df.describe()
16
17 avgprice=df["Price"].mean()
18 print(avgprice)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#01Q3.py
  Product_ID Product_Name  Price  Category  Stock_Quantity
0          1    Shampoo  120000    Hygiene           1000
1          2  Hair Brush   2500  Personal Accesories           500
2          3      Soap    4500    Hygiene          10000
3          4    Charger   6000  Electronics           1000
4          5  Tooth Brush  35000    Hygiene           1500
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Product_ID      5 non-null     int64
1   Product_Name    5 non-null     object
2   Price           5 non-null     int64
3   Category        5 non-null     object
4   Stock_Quantity  5 non-null     int64
dtypes: int64(3), object(2)
memory usage: 332.0+ bytes
33600.0
```

EXERCISE # 02: DATA LOADING AND BASIC OPERATIONS

Q1. Load the provided employee_data.csv file into a DataFrame called df_emp.

SOLUTION:

```
1 import pandas as pd
2 df=pd.read_csv('employee.csv')
3 print(df)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#02Q1.py
```

	ID	Department	Department ID	Employee ID	Employee Name	Phone	Email	SALARY
0	1.0	IT	101.0	EMP1001	Ali Khan	0300-1000000	ali.khan@company.com	50000
1	2.0	HR	102.0	EMP1002	Ahmed Raza	0300-1000001	ahmed.raza@company.com	60000
2	3.0	Finance	103.0	EMP1003	Umer Pervez	0300-1000002	umer.pervez@company.com	50000
3	4.0	Marketing	104.0	EMP1004	Hamza Malik	0300-1000003	hamza.malik@company.com	5000
4	5.0	Sales	105.0	EMP1005	Saad Ahmed	0300-1000004	saad.ahmed@company.com	50000
5	6.0	IT	101.0	EMP1006	Bilal Hussain	0300-1000005	bilal.hussain@company.com	70000
6	7.0	HR	102.0	EMP1007	Usman Tariq	0300-1000006	usman.tariq@company.com	50000
7	8.0	Finance	103.0	EMP1008	Hassan Ali	0300-1000007	hassan.ali@company.com	50000
8	9.0	Marketing	104.0	EMP1009	Zain Sheikh	0300-1000008	zain.sheikh@company.com	50000
9	10.0	Sales	105.0	EMP1010	Ayan Shah	0300-1000009	ayan.shah@company.com	50000
10	11.0	IT	101.0	EMP1011	Fahad Iqbal	0300-1000010	fahad.iqbal@company.com	50000
11	12.0	HR	102.0	EMP1012	Danish Aslam	0300-1000011	danish.aslam@company.com	50000
12	13.0	Finance	103.0	EMP1013	Arham Siddiqui	0300-1000012	arham.siddiqui@company.com	50000
13	14.0	Marketing	104.0	EMP1014	Talha Javed	0300-1000013	talha.javed@company.com	50000
14	15.0	Sales	105.0	EMP1015	Huzaifa Khan	0300-1000014	huzaifa.khan@company.com	50000
15	16.0	IT	101.0	EMP1016	Abdullah Noor	0300-1000015	abdullah.noor@company.com	50000
16	17.0	HR	102.0	EMP1017	Muneeb Akhtar	0300-1000016	muneeb.akhtar@company.com	50000
17	18.0	Finance	103.0	EMP1018	Rayan Ahmed	0300-1000017	rayan.ahmed@company.com	50000
18	19.0	Marketing	104.0	EMP1019	Shahzaib Ali	0300-1000018	shahzaib.ali@company.com	50000
19	20.0	Sales	105.0	EMP1020	Adeel Rauf	0300-1000019	adeel.rauf@company.com	50000
20	21.0	IT	101.0	EMP1021	Waleed Khan	0300-1000020	waleed.khan@company.com	50000
21	22.0	HR	102.0	EMP1022	Owais Ahmed	0300-1000021	owais.ahmed@company.com	50000
22	23.0	Finance	103.0	EMP1023	Rizwan Malik	0300-1000022	rizwan.malik@company.com	50000
23	24.0	Marketing	104.0	EMP1024	Sameer Shah	0300-1000023	sameer.shah@company.com	50000
24	25.0	Sales	105.0	EMP1025	Ibrahim Ali	0300-1000024	ibrahim.ali@company.com	50000
25	26.0	IT	101.0	EMP1026	Kashif Raza	0300-1000025	kashif.raza@company.com	50000
26	27.0	HR	102.0	EMP1027	Saif Ullah	0300-1000026	saif.ullah@company.com	50000
27	28.0	Finance	103.0	EMP1028	Noman Tariq	0300-1000027	noman.tariq@company.com	50000
28	29.0	Marketing	104.0	EMP1029	Rehan Ahmed	0300-1000028	rehan.ahmed@company.com	50000
29	30.0	Sales	105.0	EMP1030	Ammar Khan	0300-1000029	ammam.khan@company.com	50000
30	31.0	IT	101.0	EMP1031	Arsalan Javed	0300-1000030	arsalan.javed@company.com	50000
31	32.0	HR	102.0	EMP1032	Shayan Ali	0300-1000031	shayan.ali@company.com	50000
32	33.0	Finance	103.0	EMP1033	Taha Hussain	0300-1000032	taha.hussain@company.com	50000
33	34.0	Marketing	104.0	EMP1034	Yasir Malik	0300-1000033	yasir.malik@company.com	50000
34	35.0	Sales	105.0	EMP1035	Haris Raza	0300-1000034	haris.raza@company.com	50000

35	36.0	IT	101.0	EMP1036	Daniyal Khan	0300-1000035	daniyal.khan@company.com	50000
36	37.0	HR	102.0	EMP1037	Murtaza Ahmed	0300-1000036	murtaza.ahmed@company.com	50000
37	38.0	Finance	103.0	EMP1038	Ahsan Ali	0300-1000037	ahsan.ali@company.com	50000
38	39.0	Marketing	104.0	EMP1039	Farhan Sheikh	0300-1000038	farhan.sheikh@company.com	50000
39	40.0	Sales	105.0	EMP1040	Sami Ullah	0300-1000039	sami.ullah@company.com	50000
40	41.0	IT	101.0	EMP1041	Abdul Rehman	0300-1000040	abdul.rehman@company.com	50000
41	42.0	HR	102.0	EMP1042	Asad Khan	0300-1000041	asad.khan@company.com	50000
42	43.0	Finance	103.0	EMP1043	Imran Raza	0300-1000042	imran.raza@company.com	50000
43	44.0	Marketing	104.0	EMP1044	Kamran Ali	0300-1000043	kamran.ali@company.com	50000
44	45.0	Sales	105.0	EMP1045	Junaid Ahmed	0300-1000044	junaid.ahmed@company.com	50000
45	46.0	IT	101.0	EMP1046	Awais Khan	0300-1000045	awais.khan@company.com	50000
46	47.0	HR	102.0	EMP1047	Shahmeer Ali	0300-1000046	shahmeer.ali@company.com	50000
47	48.0	Finance	103.0	EMP1048	Rafay Hussain	0300-1000047	rafay.hussain@company.com	50000
48	49.0	Marketing	104.0	EMP1049	Mohsin Raza	0300-1000048	mohsin.raza@company.com	50000
49	50.0	Sales	105.0	EMP1050	Adnan Tariq	0300-1000049	adnan.tariq@company.com	50000
50	NaN	NaN	NaN	NaN	NaN	NaN	NaN	50000

Q2. Display the first 8 rows.

SOLUTION:

```
1 import pandas as pd
2 df=pd.read_csv('employee.csv')
3 print(df.head(8))
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDs Labs\LAB#02> python Lab#02Ex#02Q2.py
```

	ID	Department	Department ID	Employee ID	Employee Name	Phone	Email	SALARY
0	1.0	IT	101.0	EMP1001	Ali Khan	0300-1000000	ali.khan@company.com	50000
1	2.0	HR	102.0	EMP1002	Ahmed Raza	0300-1000001	ahmed.raza@company.com	60000
2	3.0	Finance	103.0	EMP1003	Umer Pervez	0300-1000002	umer.pervez@company.com	50000
3	4.0	Marketing	104.0	EMP1004	Hamza Malik	0300-1000003	hamza.malik@company.com	50000
4	5.0	Sales	105.0	EMP1005	Saad Ahmed	0300-1000004	saad.ahmed@company.com	50000
5	6.0	IT	101.0	EMP1006	Bilal Hussain	0300-1000005	bilal.hussain@company.com	70000
6	7.0	HR	102.0	EMP1007	Usman Tariq	0300-1000006	usman.tariq@company.com	50000
7	8.0	Finance	103.0	EMP1008	Hassan Ali	0300-1000007	hassan.ali@company.com	50000

Q3. Display only the Name and Department columns for the entire dataset.

SOLUTION:

```
1 import pandas as pd
2 df=pd.read_csv('employee.csv')
3 print(df[['Employee Name','Department']].head())
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#02Q3.py
Employee Name Department
0      Ali Khan      IT
1    Ahmed Raza      HR
2    Umer Pervez  Finance
3   Hamza Malik  Marketing
4    Saad Ahmed    Sales
```

Q4. What is the shape of the DataFrame? How many rows and columns does it have?

SOLUTION:

```
1 import pandas as pd
2 df=pd.read_csv('employee.csv')
3 print(df.shape)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#02Q4.py
(51, 8)
```

EXERCISE # 03: FILTERING DATA

Using the df_emp DataFrame:

Q1. Find all employees in the "Engineering" department.

SOLUTION:

```
1 import pandas as pd
2 df_emp=pd.read_csv('employee.csv')
3 print(df_emp[df_emp['Department']=='Engineering'])
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#03Q1.py
Empty DataFrame
Columns: [ID, Department, Department ID, Employee ID, Employee Name, Phone, Email, SALARY]
Index: []
```

Q2. Find all employees with a salary greater than 50000.

SOLUTION:

```
1 import pandas as pd
2 df_emp=pd.read_csv('employee.csv')
3 print(df_emp[df_emp['SALARY']>50000])
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#03Q2.py
```

	ID	Department	Department ID	Employee ID	Employee Name	Phone	Email	SALARY
1	2.0	HR	102.0	EMP1002	Ahmed Raza	0300-1000001	ahmed.raza@company.com	60000
5	6.0	IT	101.0	EMP1006	Bilal Hussain	0300-1000005	bilal.hussain@company.com	70000

Q3. Find all employees in "Marketing" with a salary less than 60000.

SOLUTION:

```
1 import pandas as pd
2 df_emp=pd.read_csv('employee.csv')
3 print(df_emp[(df_emp['Department']=='Marketing') & (df_emp['SALARY']<60000)])
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTD Labs\LAB#02> python Lab#02Ex#03Q3.py
```

	ID	Department	Department ID	Employee ID	Employee Name	Phone	Email	SALARY
3	4.0	Marketing	104.0	EMP1004	Hamza Malik	0300-1000003	hamza.malik@company.com	5000
8	9.0	Marketing	104.0	EMP1009	Zain Sheikh	0300-1000008	zain.sheikh@company.com	50000
13	14.0	Marketing	104.0	EMP1014	Talha Javed	0300-1000013	talha.javed@company.com	50000
18	19.0	Marketing	104.0	EMP1019	Shahzaib Ali	0300-1000018	shahzaib.ali@company.com	50000
23	24.0	Marketing	104.0	EMP1024	Sameer Shah	0300-1000023	sameer.shah@company.com	50000
28	29.0	Marketing	104.0	EMP1029	Rehan Ahmed	0300-1000028	rehan.ahmed@company.com	50000
33	34.0	Marketing	104.0	EMP1034	Yasir Malik	0300-1000033	yasir.malik@company.com	50000
38	39.0	Marketing	104.0	EMP1039	Farhan Sheikh	0300-1000038	farhan.sheikh@company.com	50000
43	44.0	Marketing	104.0	EMP1044	Kamran Ali	0300-1000043	kamran.ali@company.com	50000
48	49.0	Marketing	104.0	EMP1049	Mohsin Raza	0300-1000048	mohsin.raza@company.com	50000

Q4. Challenge: Select all employees whose name starts with the letter 'J'. (Hint: look up the.str.startswith() method for Pandas Series).

SOLUTION:

NOTE: We use na=False because the last row has missing employee information. Without na=False, the missing value (NaN) could cause an error while filtering the DataFrame.

```
1 import pandas as pd
2 df_emp=pd.read_csv('employee.csv')
3 print(df_emp[df_emp['Employee Name'].str.startswith('J',na=False)])
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTD Labs\LAB#02> python Lab#02Ex#03Q4.py
```

	ID	Department	Department ID	Employee ID	Employee Name	Phone	Email	SALARY
44	45.0	Sales	105.0	EMP1045	Junaid Ahmed	0300-1000044	junaaid.ahmed@company.com	50000

EXERCISE # 04: HANDLING MISSING DATA

A file customer_feedback.csv has been loaded into a DataFrame df_feedback, but it's messy.

Q1. Check how many missing values exist in each column.

SOLUTION:

```
1 import pandas as pd
2 df_customerfeedback=pd.read_csv('customer_feedback.csv')
3 print(df_customerfeedback.isnull().sum())
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTD Labs\LAB#02> python Lab#02Ex#04Q1.py
customer_id      0
customer_name    0
rating           5
comment          4
dtype: int64
```

Q2. The rating column is numerical. Fill its missing values with the overall average rating.

SOLUTION:

```
1 import pandas as pd
2 df_customerfeedback=pd.read_csv('customer_feedback.csv')
3 df_customerfeedback['rating']=df_customerfeedback['rating'].fillna(df_customerfeedback['rating'].mean())
4 print(df_customerfeedback)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTD Labs\LAB#02> python Lab#02Ex#04Q2.py
customer_id customer_name rating comment
0 1 Ali 5.000000 Excellent service!
1 2 Sara 4.000000 Very good experience.
2 3 Ahmed 3.666667 Good overall.
3 4 Hina 3.000000 NaN
4 5 Usman 5.000000 Amazing customer service.
5 6 Ayesha 3.666667 Could be better.
6 7 Bilal 2.000000 Not satisfied.
7 8 Zainab 4.000000 Pretty good.
8 9 Hamza 3.000000 NaN
9 10 Mariam 3.666667 The experience was okay.
10 11 Danish 5.000000 Loved it!
11 12 Fatima 1.000000 Very poor service.
12 13 Owais 4.000000 Helpful staff.
13 14 Kiran 3.666667 Good service.
14 15 Saad 2.000000 NaN
15 16 Javeria 5.000000 Excellent!
16 17 Hassan 3.000000 Average experience.
17 18 Mahnoor 3.666667 Nice service.
18 19 Raza 4.000000 NaN
19 20 Iqra 5.000000 Highly recommended.
```

Q3. The comment column is textual. Drop any rows where the comment is missing.

SOLUTION:

```
1 import pandas as pd
2 df_customerfeedback=pd.read_csv('customer_feedback.csv')
3 df_customerfeedback=df_customerfeedback.dropna(subset=['comment'])
```

OUTPUT:

NOTE: No output was shown because we did not use `print(df_customerfeedback)`. The cleaned dataframe was shown in q4.

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#04Q3.py
```

Q4. Display the cleaned DataFrame to confirm your changes.

SOLUTION:

```
1 import pandas as pd
2 df_customerfeedback=pd.read_csv('customer_feedback.csv')
3 df_customerfeedback=df_customerfeedback.dropna(subset=['comment'])
4 print(df_customerfeedback)
```

OUTPUT:

```
PS C:\Users\USER\Desktop\ITTDS Labs\LAB#02> python Lab#02Ex#04Q4.py
```

	customer_id	customer_name	rating	comment
0	1	Ali	5.0	Excellent service!
1	2	Sara	4.0	Very good experience.
2	3	Ahmed	NaN	Good overall.
4	5	Usman	5.0	Amazing customer service.
5	6	Ayesha	NaN	Could be better.
6	7	Bilal	2.0	Not satisfied.
7	8	Zainab	4.0	Pretty good.
9	10	Mariam	NaN	The experience was okay.
10	11	Danish	5.0	Loved it!
11	12	Fatima	1.0	Very poor service.
12	13	Owais	4.0	Helpful staff.
13	14	Kiran	NaN	Good service.
15	16	Javeria	5.0	Excellent!
16	17	Hassan	3.0	Average experience.
17	18	Mahnoor	NaN	Nice service.
19	20	Iqra	5.0	Highly recommended.